F12 4th CAR

Lecture 35: Section 5.4
Sum and Difference Formulas

Sum and difference formulas for sine, cosine, and tangent

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Sum and Difference Formulas

memorize

Formulas for sine:

$$\sin(u+v) = \sin u \cos v + \cos u \sin v$$

$$\sin(u-v) = \sin u \cos v - \cos u \sin v$$

Formulas for cosine:

$$\cos(u+v) = \cos u \cos v - \sin u \sin v$$

$$\cos(u-v) = \cos u \cos v + \sin u \sin v$$

Formulas for tangent:

$$\tan(u+v) = \frac{\tan u + \tan v}{1 - \tan u \tan v}$$

$$\tan(u-v) = \frac{\tan u - \tan v}{1 + \tan u \tan v}$$

ex. Find the exact value.

1) $\sin 75^\circ = \sin(30^\circ + 45^\circ) =$

$= \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ$

$= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \boxed{\frac{\sqrt{2}+\sqrt{6}}{4}}$

$\frac{\pi}{12} = \frac{180^\circ}{12} = 15^\circ = 60^\circ - 45^\circ$

$= \frac{\pi}{3} - \frac{\pi}{4}$

2) $\cos \frac{\pi}{12} = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) =$

$= \cos\frac{\pi}{3} \cos\frac{\pi}{4} + \sin\frac{\pi}{3} \sin\frac{\pi}{4}$

$= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \boxed{\frac{\sqrt{2}+\sqrt{6}}{4}}$

3) $\tan 15° = \tan(60° - 45°) = \dfrac{\tan 60° - \tan 45°}{1 + \tan 60° \tan 45°}$

$= \dfrac{\sqrt{3} - 1}{1 + \sqrt{3}(1)} = \left(\dfrac{\sqrt{3} - 1}{1 + \sqrt{3}}\right)\left(\dfrac{1 - \sqrt{3}}{1 - \sqrt{3}}\right) = \dfrac{\sqrt{3} - 3 - 1 + \sqrt{3}}{1 - 3} =$

$= \dfrac{-\frac{1}{2}}{-\frac{1}{2}}\left(\dfrac{2\sqrt{3} - 4}{-2}\right) = \dfrac{-\sqrt{3} + 2}{1} = \boxed{2 - \sqrt{3}}$

ex. Find the exact value.

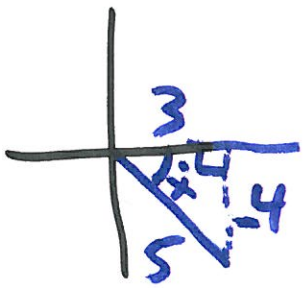
1) $\sin 20° \cos 40° + \cos 20° \sin 40° = \sin(20° + 40°)$

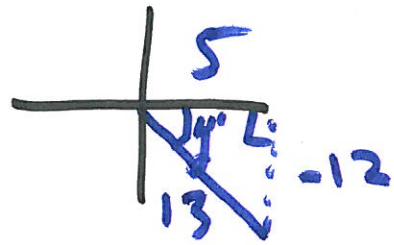
$= \sin 60° = \boxed{\dfrac{\sqrt{3}}{2}}$

2) $\dfrac{\tan 73° - \tan 13°}{1 + \tan 73° \tan 13°} = \tan(73° - 13°) = \tan 60°$

$= \boxed{\sqrt{3}}$

ex. If $\sin x = -\frac{4}{5}$, $\cos y = \frac{5}{13}$ and both x and y are in quadrant IV, find the exact value of





1) $\sin(x-y) = \sin x \cos y - \cos x \sin y$

$$\left(\frac{-4}{5}\right)\left(\frac{5}{13}\right) - \left(\frac{3}{5}\right)\left(\frac{-12}{13}\right) =$$

$$= \frac{-20+36}{5 \cdot 13} = \boxed{\frac{16}{65}}$$

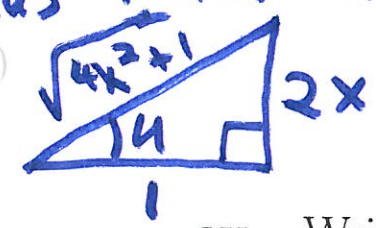
2) $\cot(x+y) = \dfrac{1}{\tan(x+y)} = \dfrac{1}{\frac{56}{33}} = \boxed{\dfrac{33}{56}}$

$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y} = \frac{\frac{-4}{3} + \frac{-12}{5}}{1 - \left(\frac{-4}{3}\right)\left(\frac{-12}{5}\right)} = \frac{56}{33}$$

flip answer

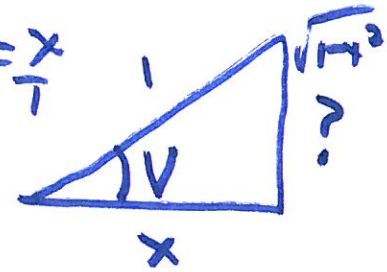
$u = \arctan 2x$

thus $\tan u = 2x$



$v = \arccos x$

thus $\cos v = x = \frac{x}{1}$



ex. Write $\sin(\overbrace{\arctan 2x}^{u} - \overbrace{\arccos x}^{v})$ as an algebraic expression.

$x^2 + ?^2 = 1^2$

$?^2 = 1 - x^2$

$? = \pm\sqrt{1-x^2}$

$=$

$$\sin(u-v) = \sin u \cos v - \cos u \sin v$$

$$= \left(\frac{2x}{\sqrt{4x^2+1}}\right)\left(\frac{x}{1}\right) - \frac{1}{\sqrt{4x^2+1}}\left(\frac{\sqrt{1-x^2}}{1}\right)$$

$$= \boxed{\frac{2x^2 - \sqrt{1-x^2}}{\sqrt{4x^2+1}}}$$

ex. Prove the cofunction identity $\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$.

$$LHS = \sin\left(\frac{\pi}{2} - \theta\right) = \sin\frac{\pi}{2}\cos\theta - \cos\frac{\pi}{2}\sin\theta$$

$$= 1 \cdot \cos\theta - 0 \cdot \sin\theta$$

$$= \cos\theta = RHS \checkmark$$

ex. Verify $\cos(\theta + 2\pi) = \cos\theta$.

$$LHS = \cos(\theta + 2\pi) = \cos\theta\cos 2\pi - \sin\theta\sin 2\pi$$

$$= \cos\theta \cdot 1 - \sin\theta \cdot 0$$

$$= \cos\theta = RHS \checkmark$$

ex. Verify the identity $\dfrac{1 + \tan x}{1 - \tan x} = \tan\left(\frac{\pi}{4} + x\right)$.

$$RHS = \tan\left(\frac{\pi}{4} + x\right) = \frac{\tan\frac{\pi}{4} + \tan x}{1 - \tan\frac{\pi}{4}\tan x} = \frac{1 + \tan x}{1 - 1(\tan x)}$$

$$= \frac{1 + \tan x}{1 - \tan x} = LHS \checkmark$$

$f(x) = \cos x$

$f(x+h) = \cos(x+h) = \text{etc.}$

<u>ex.</u> If $f(x) = \cos x$, verify the identity used in calculus:

$$\frac{f(x+h) - f(x)}{h} = \cos x \left(\frac{\cos h - 1}{h}\right) - \sin x \left(\frac{\sin h}{h}\right)$$

where $h \neq 0$.

$$LHS = \frac{\cos(x+h) - \cos x}{h} =$$

$$= \frac{\cos x \cos h - \sin x \sin h - \cos x}{h}$$

$$= \frac{\cos x \cos h - \cos x - \sin x \sin h}{h}$$

$$= \frac{\cos x \cos h - \cos x}{h} - \frac{\sin x \sin h}{h}$$

$$= \frac{\cos x(\cos h - 1)}{h} - \sin x\left(\frac{\sin h}{h}\right) = RHS$$

Solving a Trigonometric Equation

<u>ex.</u> Find all solutions of

$$\sin\left(x+\frac{\pi}{6}\right)-\sin\left(x-\frac{\pi}{6}\right)=\frac{1}{2}$$

in the interval $[0, 2\pi)$.

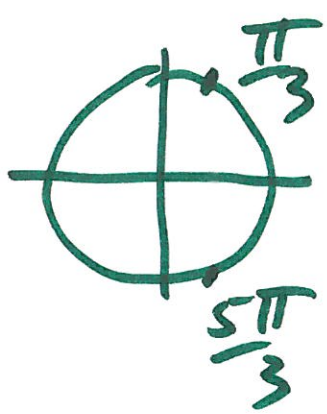
$$\sin x\cos\frac{\pi}{6}+\cos x\sin\frac{\pi}{6}-\left(\sin x\cos\frac{\pi}{6}-\cos x\sin\frac{\pi}{6}\right)=\frac{1}{2}$$

$$\cos x\sin\frac{\pi}{6}+\cos x\sin\frac{\pi}{6}=\frac{1}{2}$$

$$\cos x\left(\frac{1}{2}\right)+\cos x\left(\frac{1}{2}\right)=\frac{1}{2}$$

$$\frac{1}{2}\cos x+\frac{1}{2}\cos x=\frac{1}{2}$$

$$\cos x=\frac{1}{2}$$



$$\boxed{x=\frac{\pi}{3},\ \frac{5\pi}{3}}$$